

Evaluating K–12 AI Learning Modules: Standards Alignment and the Hidden Curriculum of AI Education

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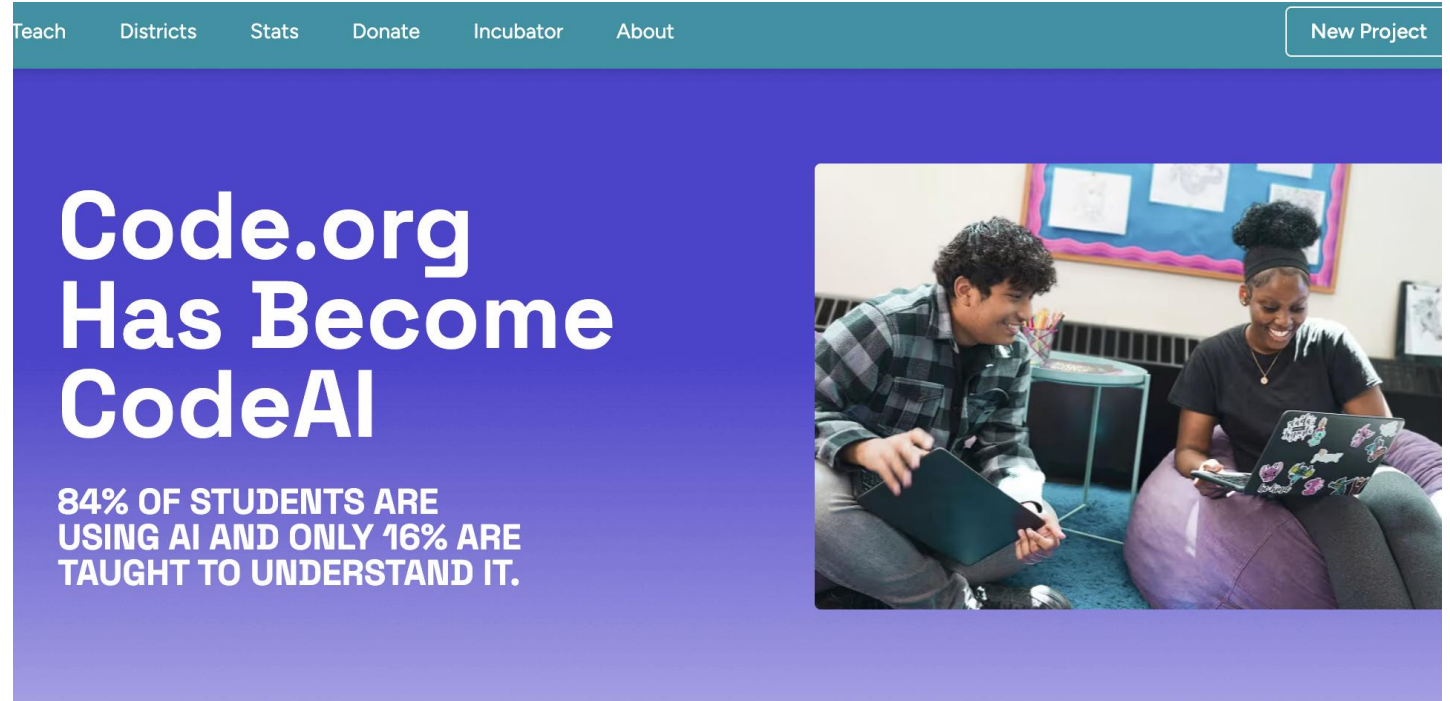


Background

- AI literacy is critical for K-12 education to prepare future citizens and strengthen national competitiveness
- Challenges faced by teachers:
 - Limited understanding of AI concepts
 - Lack of professional development opportunities
 - Limited AI literacy curriculum guidance

Code.org

- Provides accessible AI and computing resources for K-12 teachers
- Lesson plans and interactive activities including AI learning games
 - Ex: “How AI Makes Decisions” module for grades 3-5



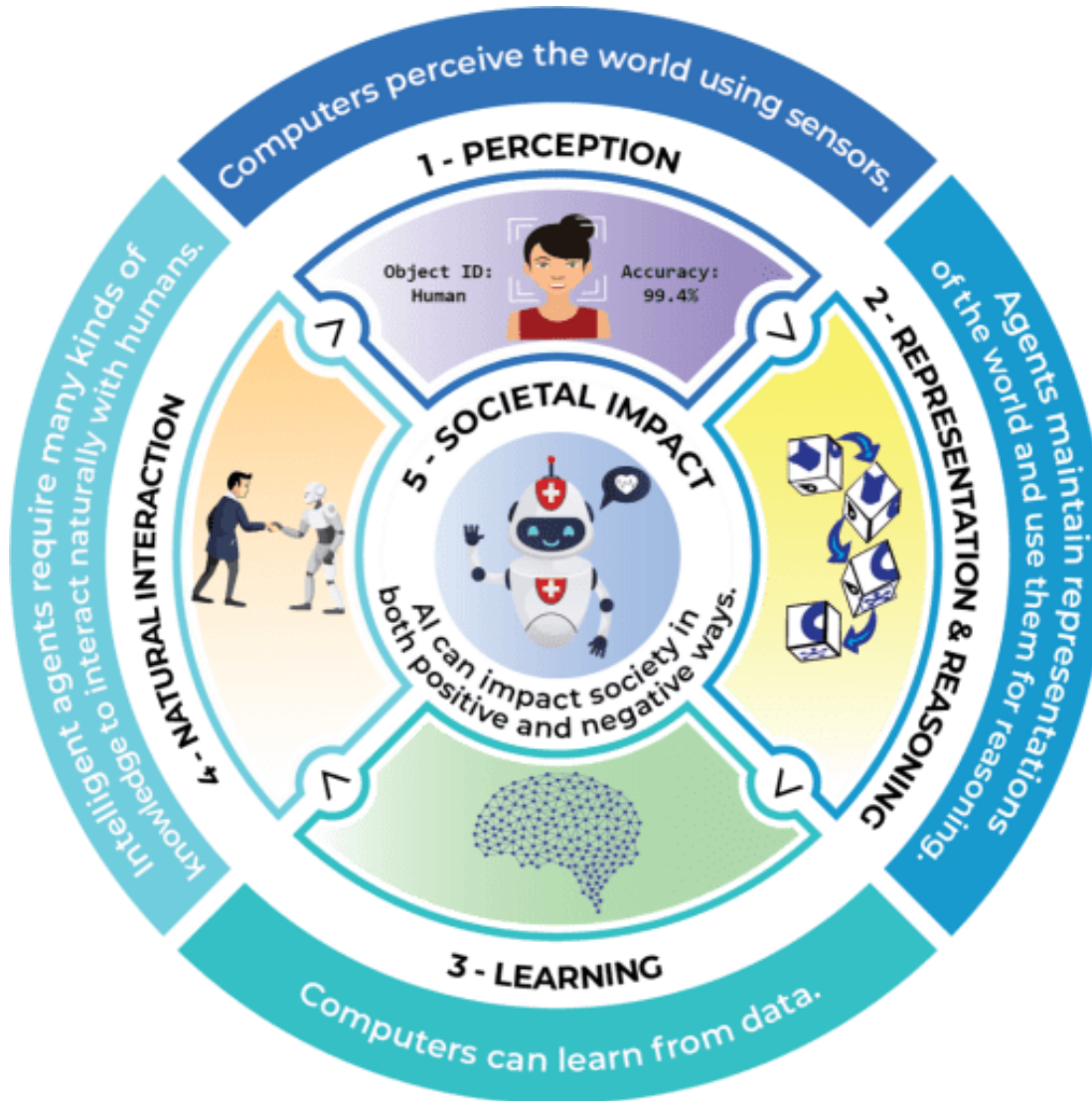
“We Have A Responsibility To Prepare The Next Generation For The Biggest Change In Society Since The Invention Of Public Education.”

AI4K12 Guidelines

➤ Artificial Intelligence for K-12 (AI4K12) initiative

➤ 5 Big Ideas:

1. Perception
2. Representation and Reasoning
3. Learning
4. Natural Interaction
5. Societal Impact



(Grade Band Progression Charts, 2021)

Purpose

To what extent and in what ways the AI learning modules from Code.org teach students literacy for critically engaging with AI tools.

RQ1

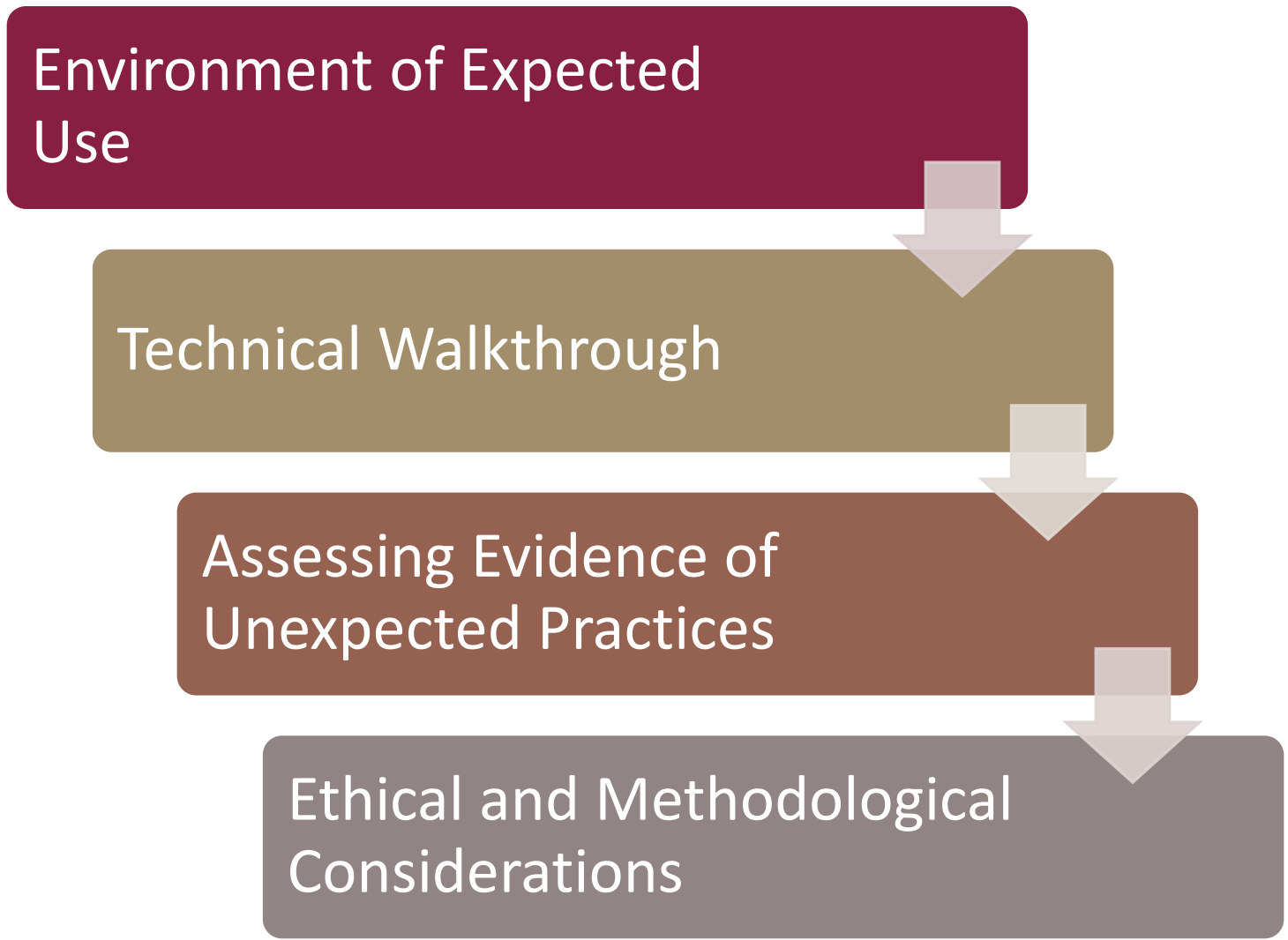
To what extent do the lessons in the module align with or diverge from AI4K12 guidelines?

RQ2

What ethical implications and values about AI in society are communicated through the module?

Methods

Environment of Expected
Use



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graph TD; A[Environment of Expected Use] --> B[Technical Walkthrough]; B --> C[Assessing Evidence of Unexpected Practices]; C --> D[Ethical and Methodological Considerations];
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The diagram illustrates a four-step process for the Walkthrough Method. It consists of four rectangular boxes arranged in a descending staircase pattern from top-left to bottom-right. Each box is connected to the next by a downward-pointing arrow. The boxes are colored in a gradient: dark red, olive green, brown, and grey.

Technical Walkthrough

Assessing Evidence of
Unexpected Practices

Ethical and Methodological
Considerations

Walkthrough Method

(Light et al., 2018)

Environment of Expected Use

- Context of the app
 - AI module's lesson plans
 - Information pages from Code.org
- Analyze in conjunction with technical walkthrough data

Technical Walkthrough

LIGHT ET AL. TECHNICAL WALKTHROUGH

- Alignment between app functionalities and design principles
- Ex: Usability

OUR TECHNICAL WALKTHROUGH

- Alignment between app experience and learning standards
- Ex: Experience with learning the AI4K12 5 Big Ideas

Lesson	Standard(s)	Activity	Learning activity descriptions/reference images	Values about AI Literacy (activities) - What is the learning activity teaching students about how AI works?	Alignment - To what extent does the lesson meet the standard(s)? - Alignments and misalignments between standard(s) and the lesson	Values about AI Literacy (lesson) - What is the lesson teaching students about the societal impact of AI?

Assessing Evidence of Unexpected Practices

RQ1. To what extent do the lessons in the module align with or diverge from AI4K12 guidelines?

- “Alignment” column of the workbook
- Alignment and misalignment of “How AI Makes Decisions” module and AI4K12 standards
- Curriculum Alignment theory (Leitzel & Vogler, 1994)

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Ethical and Value Implications

RQ2. What ethical implications and values about AI in society are communicated through the module?

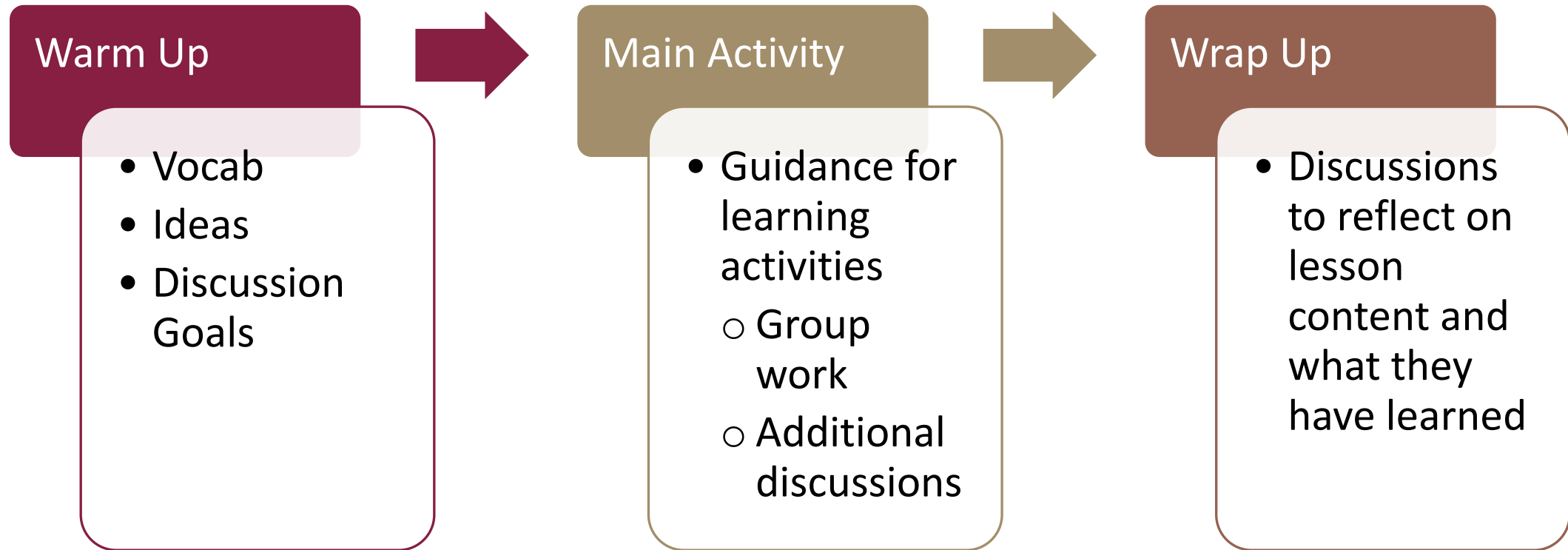
- Environment of expected use
- “Values about AI Literacy” columns of the workbook
- Hidden curriculum theory (Jackson, 1968)

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Findings

ENVIRONMENT OF EXPECTED USE

AI Module “Vision”



Code.org “Operating Model”

- Small donations from individuals
- Large donations from corporations and other sources
 - Donors do not have control over spending or curriculum design
 - No access to student data
- Code.org does not sell user data

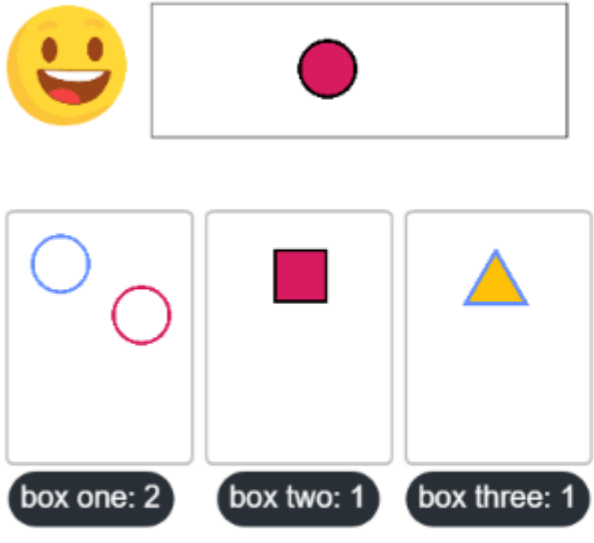
Code.org “Governance”

- Code.org curriculum and tutorials can be used and adapted by educators for non-commercial purposes
- Little governance
- Code.org aims for their curriculum to be politically and morally neutral

Findings

THE TECHNICAL WALKTHROUGH

Lesson 1: Making Predictions



box one: 2 box two: 1 box three: 1

Run

Instructions

Look at the items on the left. Predict where the circle with the black border and pink color will go.

- ☐ A. **Box One** sorted by shape
- ☐ B. **Box Two** sorted by inside color
- ☐ C. **Box Three** sorted by border color
- ☐ D. It could go in **Box One** or **Box Two**. We need more training data to know.

Lesson 1, Activity 3

Lesson 1: Making Predictions

3-A-i. Nature of Learning (Humans vs. Machines)

- Partial alignment
- Compares how humans and machines learn, but does not contrast

3-A-iii. Nature of Learning (Training a model)

- One researcher said no alignment
- Two researchers said some alignment
- Lesson introduces basic ideas of data labeling and training data

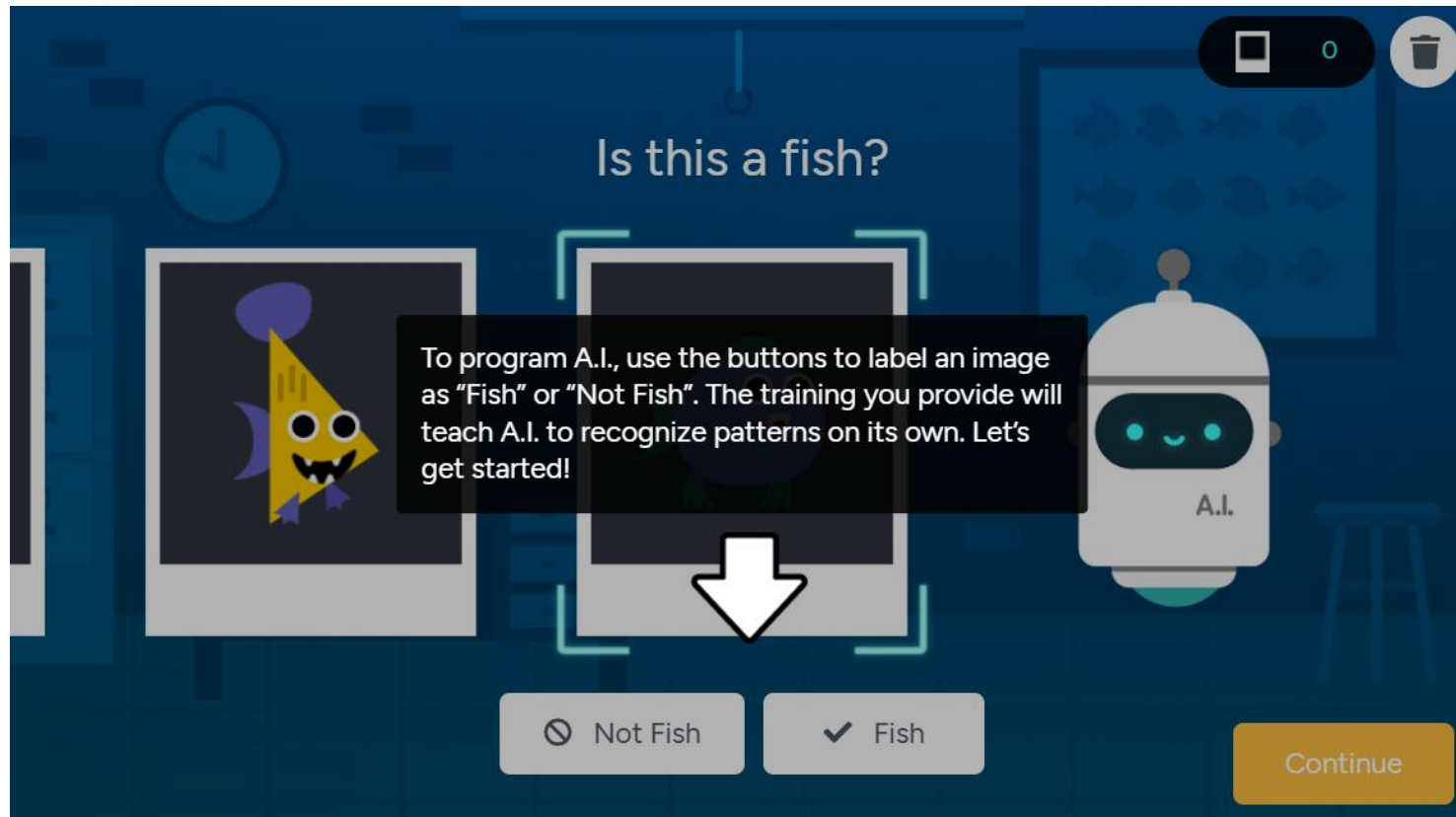
5-B-i. AI & Culture (AI in Daily Life)

- No examples of AI-powered services in daily life

Hidden Curriculum

- Frames AI as a technical sorting process.
- Omits discussion of ethics, bias, and societal impacts.
- Implicitly presents AI as a neutral, value-free technology.
- Lacks real-world examples that support ethical reflection.

Lesson 2: Training AI



Lesson 2, Activity 2

Lesson 2: Training AI

3-A-i. Nature of Learning (Humans vs. Machines)

- Partial alignment
- “Machine Learning Video” activity aligns significantly
- Compares how humans and machines learn, but does not contrast

3-A-iii. Nature of Learning (Training a model)

- Well-aligned with the standard

5-B-i. AI & Culture (AI in Daily Life)

- No examples of AI-powered services in daily life
- AI being used to clean garbage from the ocean is a real-world example

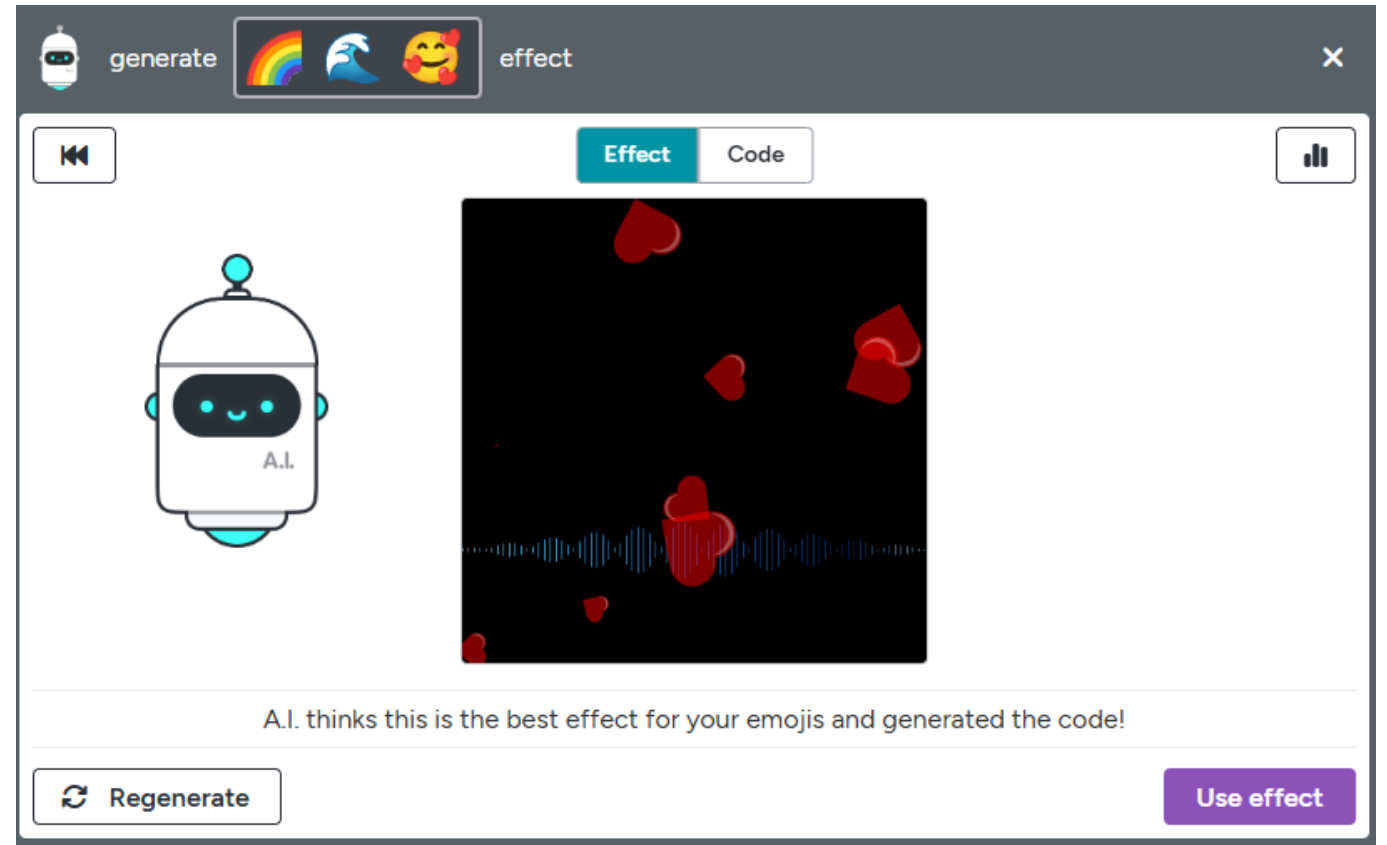
Hidden Curriculum

- Situates AI in a socially relevant context (e.g., ocean cleanup).
- Introduces concepts of fairness, bias, and unintended consequences.
- Highlights that AI decisions are shaped by human training choices.
- Focuses on technical improvement through retraining and accuracy.
- Provides limited discussion of broader societal impacts of AI.

Lesson 3: Using AI Bot



Lesson 3, Activity 3



Lesson 3, Activity 6

Lesson 3: Using AI Bot

2-A-iv. Representation (Feature vectors)

- One researcher said strong alignment
- Two researchers said activities 1 and 3 were somewhat aligned

3-A-i. Nature of Learning (Humans vs. Machines)

- “Video: How AI Works” activity somewhat aligns

5-B-i. AI & Culture (AI in Daily Life)

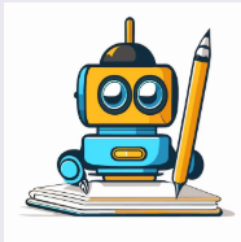
- “Video: How AI Works” activity is well-aligned
- One researcher felt activities 2 and 3 also aligned

Hidden Curriculum

- Presents AI as a tool for identifying patterns and associations from data.
- Emphasizes prediction, recommendation, and content generation.
- Frames AI-generated outputs as useful and actionable.
- Provides limited opportunities to evaluate the validity, trustworthiness, or societal consequences of AI outputs.

Lesson 4: AI Tools in School

Your school is considering using an A.I. tool for the following scenarios. Choose one:

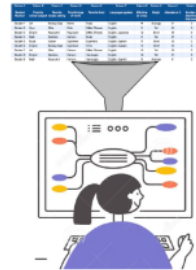


- In the cafeteria, students are seated alphabetically by last name.
The Lunch Aide wants to use AI to seat students based on their interests.
- Every month, there is a new Student of the Month.
The 3rd grade teacher wants to use AI to help him choose this month's student.
- The gym teacher usually chooses teams randomly.
Now she wants to use AI to help her form teams.

Then look at the [Student Dataset](#) and answer the questions below:



Step 1: Choose the data



Step 2: Give the computer the training data

Lesson 4, Activity 2

Lesson 4: AI Tools in School

3-A-i. Nature of Learning (Humans vs. Machines)

- Two researchers said no alignment
- One researcher said activity two is somewhat aligned

3-C-iii. Datasets (Bias)

- Partially aligned
- Covered training data and datasets, but not bias

5-B-i. AI & Culture (AI in Daily Life)

- Partially aligned
- Real-world training data and realistic scenarios
- No AI-powered systems used in daily life

Hidden Curriculum

- Demonstrates how AI can be used to address real-world school problems.
- Emphasizes the importance of selecting relevant training data.
- Presents AI as a useful tool for decision-making.
- Implicitly frames social problems as solvable through data and AI design.
- Provides limited discussion of bias, limitations, risks, and responsibility.

Lesson 5: Share Your Voice

Choose from the following activities:

	a Letter to Your Principal 👉 Share your thoughts on AI by writing a letter to your principal.		b Make a Poster 👉 Share your thoughts on AI by making a hallway poster.		c Share Your Voice 👉👉 Share your thoughts on AI by making a speech, writing a rap, or making up a song.
	d Create a Sprite Lab project 👉👉 Share your thoughts on AI by coding a Sprite Lab project.	e Code a Poem 👉👉👉 Share your thoughts on AI by coding a poem.			

Back

Finish

Lesson 5, Activity 1

Lesson 5: Share Your Voice

3-A-i. Nature of Learning (Humans vs. Machines)

3-C-iii. Datasets (Bias)

5-B-i. AI & Culture (AI in Daily Life)

Hidden Curriculum

- Action-oriented projects (e.g., letters, posters, speeches, songs) encourage reflection on AI's benefits and drawbacks.
- Students consider how AI may be used in school settings.
- Coding projects focus primarily on technical implementation.
- Ethical reflection is emphasized unevenly across project options. Implicitly positions ethics as separate from technical design.

Cross-Lesson Findings and Implications

- Lessons were generally stronger in teaching technical AI concepts than broader AI literacy competencies.
- Alignment with AI4K12 standards was often partial rather than comprehensive.
- AI was frequently presented as a useful tool for solving problems, making predictions, and generating outputs.
- Ethical and societal issues were present, but were often embedded within activities and discussions rather than highlighted as central learning goals.
- Hidden curriculum messages emphasized building, training, and improving AI more than critically examining its assumptions, limitations, and social implications.
- Ethical reflection was often positioned accompanying AI applications rather than fully integrated into technical design activities.

Discussion: Why Do These Patterns Appear?

- **Developmentally Appropriate Simplification**

- Complex AI ethics and societal issues were translated into concrete, age-appropriate examples and activities.
- This may make ethical concerns more abstract and less connected to real-world sociotechnical challenges.

- **Commitment to Political and Moral Neutrality**

- Code.org explicitly aims to provide politically and morally neutral curricula.
- Ethical issues may therefore be framed through technical concepts such as bias, accuracy, and data quality rather than normative ethical frameworks.

- **Accessibility and Broad Adaptability**

- The curriculum is designed for diverse educational settings and learner populations.
- Ethical discussions are often embedded in teaching guides and classroom discussions, allowing teachers flexibility to adapt content to local contexts.

- **Foundational AI Literacy Focus**

- The primary goal appears to be introducing core AI concepts and skills.
- As a result, technical understanding often receives greater emphasis than critical examination of AI's social and ethical implications.

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